

Introduction to Uncertainty and Probabilistic Reasoning

Video Lesson

Duration: 00:07:31

Slide Deck

M11S01 (<https://canvas.tamu.edu/courses/430127/files/92570687?wrap=1>) 

Lesson Transcript

In this lecture, we are looking into uncertainty and probabilistic reasoning.

Uncertainty

- Problem with first-order logic: agents almost never have full access to the whole truth about their environment.
- Therefore, the agent must act under **uncertainty**.
- Uncertainty can also arise because of incompleteness and incorrectness in the agent's understanding of the properties in the environment.
- Incomplete, because there are too many conditions to explicitly enumerate.

There are trade-offs (playing safe can result in other annoyances), thus the right thing to do depends on both the relative importance of various goals and the likelihood (and degree to which) they will be achieved.

Uncertainty can be a huge issue for logic-based AI systems. One of the problems with first-order logic is that agents almost never have full access to the whole truth about their environment. Due to this, the agent must act under some kind of uncertainty.

Uncertainty can also arise because of incompleteness and incorrectness in the agent's understanding of the properties in the environment. It is incomplete because there are too many conditions to enumerate explicitly.

In this kind of situation, you can try to play safe, but this can result in other annoyances. Hence, there is a trade-off. In this kind of situation, the right thing to do depends on both the relative importance of various goals and the likelihood and degree to which they will actually be achieved.

Example: Trying to Catch a Flight

A_t : plan to leave home t minutes before the flight departure time.

- The traveler needs to make a decision in an uncertain environment: car can break down, traffic can be extremely congested, natural disaster, etc.
- Such worst-case scenarios are hard to explicitly enumerate: the list goes on – ran out of gas, spouse/children in an emergency, flight crews goes on a strike, etc. etc.
- Thus the traveler only has an incomplete understanding of the situation.
- The traveler can play safe by going with plan A_{1440} , but this can cause the traveler to wait for a long time at the airport before departure.

For example, consider the task of trying to catch a flight. And let us say you have a plan A_t . This

plan says we plan to leave home t minutes before the flight departure time.

For the College Station airport, maybe this would be around 90 minutes. For Houston, it may be over 200 minutes. And so on. Aside from the distance to the airport, several different factors would affect our decision on how many minutes ahead of time you want to leave your home. And there are lots of uncertainties in this environment.

For example, your car can break breakdown, or there could be unexpected heavy traffic or a natural disaster such as flooding.

These worst-case scenarios are very hard to explicitly enumerate because the list can go on. For example, you can run out of gas, your family may have an emergency, or the flight crews may go on strike, and so on.

As a result, the traveler only has an incomplete understanding of the situation. The traveler can decide to play safe by going with plan A_{1440} . That is actually 24 hours, but this will be a major inconvenience.

Difficulties in Applying F-O-L in Uncertain Domains

For example, application of first-order logic in medical diagnosis domain can fail because of these reasons:

- Laziness: cannot list the complete set of antecedents and consequents needed to ensure an exceptionless rule, and too hard to use the enormous rules that result.
- Theoretical ignorance: medical science has no complete theory.
- Practical ignorance: even though we have all the rules, it is practically impossible to run all the tests.

Similar situation arises in law, business, dating, etc. The agent's knowledge can at best provide only a **degree of belief**. **Probability theory** is well suited for such a domain.

There are several difficulties in applying first-order logic in this kind of uncertain domain. Let us consider the medical diagnosis domain. Application of first-order logic in this kind of domain can fail for three reasons. These are laziness, theoretical ignorance, and practical ignorance.

Laziness says we cannot list the complete set of antecedents and consequents needed to ensure an exceptionless rule. And it is too hard to use the enormous rules that result.

Theoretical ignorance says medical science has no complete theory.

As for practical ignorance, it says even though we have all the rules, it is practically impossible to

run all the tests that are required to disambiguate any cases.

Similar situations arise in law, business, dating, and so on. The agent's knowledge can, at best, provide only a degree of belief, not a definite true or false, and probability theory is well suited for such a domain.

Acquisition of New Information and Probability_I

- The degree of belief changes as an agent perceives or acquires new information from the world: we call this the **evidence**.
- This is analogous to saying whether or not a given logical sentence is entailed by (i.e. is a logical consequence of) the knowledge base, because the truth value can change when new facts are added to the KB.
- Before the evidence is received, we talk about **prior** or **unconditional** probability.
- After the evidence is obtained, we talk about **posterior** or **conditional** probability.

As we saw in the knowledge base case, when new information is given, new facts can be added to the knowledge base.

So, what about this probabilistic domain? What happens in this domain is that the degree of belief changes as an agent proceeds or acquires new information from the world. And we call this the evidence.

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$$P(\text{Rain}) = 0.2$$
$$P(\text{Rain} \mid \underbrace{\text{Wet Umbrella}}_{\text{evidence}}) = 0.9$$

Let us consider this probability, the probability of it raining outside.

If this were for any arbitrary day, then maybe the probability would be pretty low in Texas.



Let us say that the probability is 0.2.

Now, suppose you received this new evidence, a wet umbrella. For example, if someone walked into the room with a wet umbrella, then would your belief about it raining outside change? Would it go up, or would it go down?

Normally this would actually go up because the wet umbrella is a pretty good indicator that it is already raining heavily outside.

This is kind of similar to saying whether or not a given logical sentence is entailed by the knowledge base because the truth value can change when new facts are added to the knowledge base.

Before the evidence is received, we talk about prior or unconditional probability, and after the evidence is obtained, we talk about posterior or conditional probability.

So, the probability of rain is a prior probability, and the probability of rain given a wet umbrella is a posterior probability or conditional probability.

Example

When playing black jack,

- as new cards are drawn and shown, your degree of belief in the fact that you need more cards can change.

What about poker? or slot machine?

$$P(\text{Win}) = 0.499$$

$$P(\text{Win} \mid \boxed{\diamond^a}, \boxed{\spadesuit^k}) = 0.99$$

Consider the card game Blackjack. If you get a total sum of 21, you win. Now we can think about the prior probability. What is the probability of winning?

Usually, this is close to 0.5 in any gambling situation, but there is some edge to the house. Let us say the probability of winning without any information is 0.499. After receiving two cards, one is an ace, and one is a king, which sums up to 21. Then we can say my chance of winning is very high, 0.99. Of course, it may not be this high because there is a chance of drawing with the dealer. Anyway, as new cards are drawn and shown, your degree of belief in the fact that you will need more cards can change depending on your chance of winning.

So, if you think your chance of winning is high, then you do not need any more cards.

Example

When playing black jack,

- as new cards are drawn and shown, your degree of belief in the fact that you need more cards can change.

What about poker? or slot machine?

This applies to all these gambling games like poker or slot machines. You have to decide whether to draw more cards or drop or pull the lever or whatever, based on the accumulating information.

In all of these tasks, we have to make a decision in an uncertain environment, and that is the subject of the next lecture on decision theory.

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MODULE 11 |

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& PROBABILISTIC REASONING